

DC: Clinical Application of Immersive VR in Spatial Cognition: The Assessment of Spatial Memory and Unilateral Spatial Neglect in Neurological Patients

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ABSTRACT

Visual-spatial impairments and associated cognitive functions are hard to examine by traditional neuropsychological tests, but they have high potential to be examined in an ecologically valid way through the simulation of spatial information provided by immersive Virtual Reality (VR). This PhD research proposes two studies to investigate the clinical applicability and construct validity of immersive VR in neuropsychological rehabilitation: 1) The Immersive Virtual Memory Task (imVMT) was developed to examine spatial memory in a broad range of neurological patients. The main purpose was to apply a gesture-based natural hand interaction and to see in how far typical handicaps such as hemiparesis or visual field defects interfere with task performance. On a neuropsychological level, we aim to identify VR parameters which reveal and differentiate underlying cognitive sub-processes. 2) A virtual road crossing task (iVRoad) was developed to detect discrete symptoms of unilateral spatial neglect (USN) in right-hemispheric post-stroke patients. The aims are a) to externally validate iVRoad using conventional neuropsychological tests, b) to identify and evaluate relevant behavioral and eye tracking parameters to distinguish post-stroke patients with and without USN.

Index Terms: Spatial Memory—Neglect—Eye Tracking—VR.

1 INTRODUCTION

Over the past decades, VR has become a promising research tool in cognitive neuroscience [10]. Immersive VR offers a multitude of advantages for the application in clinical neuropsychology and cognitive (neuro)science. It brings high potential for the clinical practice, as human behavior can be precisely measured in safe, controlled yet realistic and dynamic environments. Through the multidimensional nature of the stimuli, VR scenarios resembling activities of daily life are useful in supporting the assessment of cognitive functions and underlying networks. Along with the increased experimental control and motivation, virtual environments can be used to engage the user in simulations that support the clinical assessment of cognitive functions. Hence, VR has become a promising tool in many research areas [11]. To fully exploit its potential for clinical research and practice, it is essential to determine which outcome parameters are most suitable for the evaluation of cognitive functions and underlying sub-processes. Further information is needed how the typical handicaps of neurological diseases can be taken into account in the development of clinical VR applications.

The general feasibility of immersive VR in neurological patients has been demonstrated [11]. However, it remains unclear whether the underlying constructs are specifically and validly captured. The first aim of this research is to (1) determine the clinical applicability

of the imVMT, a VR task to assess spatial memory, in a broad range of neurological patients, (2) estimate the extent to which immersion, presence, usability, motivational aspects, and side effects affect user experience, and (3) identify valid outcome parameters to reflect cognitive deficits and underlying sub-processes. The second part aims at a sensitive assessment of discrete symptoms of USN. This is a neuropsychological condition, in which subjects fail to detect, orient to and respond to stimuli located contralaterally to their (most often right) hemispheric lesion [2].

2 IMMERSIVE VIRTUAL MEMORY TASK

In the VRReha project “Virtual Worlds for Digital Diagnostics and Cognitive Rehabilitation”, an interdisciplinary research team examines the clinical applicability of immersive VR scenarios in individuals with different neurological impairments (e.g., stroke, traumatic brain injury). The consortium, comprising clinicians, psychologists, neuroscientists, neurologist, and software engineers, jointly develops and evaluates realistic diagnostic as well as individually tailored immersive training scenarios that resemble patients’ everyday lives to assess and train spatial memory [6, 12].



Figure 1: A subject grabs a virtual object and places it back in its original location by gesture-based natural hand interaction (imVMT).

One branch of the project focuses on the cognitive sub-processes underlying spatial memory, which is a fundamental cognitive function for storing and retrieving spatial information crucial for independent and functional living. Spatial memory deficits occur early in Alzheimer’s and in a number of other neurological diseases [1]. Therefore, measures to sensitively detect deficits in spatial memory are of high clinical relevance.

In the imVMT, subjects are to memorize and reconstruct the position of a varying number of everyday objects on a virtual table using Oculus Rift® VR glasses. Based on Leap Motion-based (i.e., controller-free) gesture detection, we aimed at natural interaction with the virtual stimuli. We analyzed precise recordings of performance (e.g., mean displacement, deviation angle, placement time) to infer underlying processes of spatial memory. To increase spatial requirements, subjects are either positioned on the original side of the table or rotated 90, 180 or 270 degrees in space. External validity will be examined with respect to a number of neuropsychological

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tests and specially designed questionnaires will be used to determine feasibility, cybersickness, immersion, presence, usability, and motivation.

3 IMMERSIVE VIRTUAL ROAD TASK

USN is one of the strongest predictors of poor functional recovery leading to more prolonged hospitalization and limiting subjects' independence in everyday life [7]. Affected individuals exhibit a reduced (visual) exploration behavior towards the contralesional side, leading to difficulties of activities of daily living and reduced life satisfaction [4]. While the symptoms are initially often visible in clinical observations, it becomes increasingly difficult to operationalize discrete symptoms reliably over the course of time [5]. Conventional neuropsychological tests contain mainly abstract tasks (e.g., line bisection), are performed in peripersonal space only, and use static, 2D methods [8]. Especially in the chronic phase, traditional tests frequently miss discrete USN symptoms, which has serious consequences in individual's everyday life [9].



Figure 2: In iVRoad the user crosses a busy street from first-person perspective to insert a letter into a mailbox.

Therefore, the realistic virtual street crossing scenario iVRoad has been developed matching the clinical requirements for the detection of discrete symptoms in USN. The virtual environment consists of two approximately 600 m long parallel two-lane, busy roads in a fictional urban environment. The user is positioned on the sidewalk observing the scenario from a first-person perspective with the task to cross the virtual street safely. The parameters of the system (e.g., gap size, speed, distractors) can be individually adapted under completely controllable and standardized conditions. Moreover, the outcome parameters provide clinicians and researchers with immediate feedback about patients' performance.

We apply the HTC Vive Pro Eye with an integrated eye tracking system and plan to identify appropriate eye tracking parameters that indicate USN symptoms. Given that USN is characterized by an inability to pay attention to targets contralesional, the analysis of eye tracking data represents an effective method to identify behavioral aspects of USN that provide direct insights into the visual exploration behavior and performance in realistic environments [3].

4 DISCUSSION

Within the proposed PhD research two studies on immersive VR were outlined. So far, we have shown that gesture-based hand interaction in VR is applicable and well-accepted in neurological patients (imVMT). We have further developed a task with a specific focus on everyday functionality to detect discrete USN symptoms by applying simple interaction (iVRoad). The insights gained from these studies

will lead to an improved understanding of spatial cognition, a more sensitive assessment of USN and lead to an improved comprehension of the applicability of VR in neurological patients. Investigating the application of clinical VR in stroke patients holds great potential to advance the VR field of research with respect to improved usability for patients and therapists. In this way, both disciplines can learn and benefit from each other in order to design future clinical VR applications in a user-friendly and externally valid manner.

5 CONCLUSION

To conclude, we expect to demonstrate that immersive VR can be used to validly assess spatial memory performance. However, to fully exploit the potential of VR in clinical application, suitable parameters need to be identified. The question remains as to how concrete cognitive processes can be precisely captured, since cognitive sub-functions in everyday tasks could overlap. It must therefore first be determined to what extent cognitive functions, including spatial cognition, can be clearly distinguished from one another amidst the multitude of collected data. A further question to be discussed in the consortium is the evaluation of combining VR and eye tracking data. In particular, we are interested in the interaction of VR, user head rotation and eye gaze with respect to the area of interest.

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